PROJECT REPORT

On

Research Paper

Submitted To

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FOR

CO-OP PROJECT AT INDUSTRY (MODULE-2)

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Semester: VIII
Session: 2025-2026

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1. Introduction

Artificial intelligence has become a regular part of modern communication. Many people now use chatbots, virtual assistants, and automated systems for guidance, support, and quick responses. These tools are helpful because they are available anytime and respond instantly. However, continuous dependence on such systems has also created new emotional challenges.

One growing issue is the feeling users experience when an AI system suddenly stops replying, becomes unavailable, or behaves unexpectedly. Since many AI platforms communicate in a human-like manner, their sudden silence can create confusion or stress. This situation is often similar to being ignored in a real conversation.

To understand this concept practically, a mini software project named **Digital Ghosting Stress Analyzer** was developed using Python and Streamlit. The system is based on the research topic and demonstrates how emotional stress caused by AI silence can be analyzed through a simple digital tool.

2. Problem Statement

AI-based communication tools are designed to assist users in daily tasks, conversations, and emotional support. But when these systems stop responding without any notice, users may feel uncomfortable, frustrated, or mentally disturbed.

At present, most AI tools focus on technical performance and speed, while the emotional side of user experience is often ignored. There are very few systems that measure the psychological effect caused by sudden AI silence.

Therefore, there is a need for a practical solution that can observe user responses and estimate stress levels related to Digital Ghosting.

3. Title of Project

Title: *"Digital Ghosting Stress Analyzer: Measuring Psychological Stress Caused by AI-Mediated Social Rejection"*

- The title **"Digital Ghosting Stress Analyzer: Measuring Psychological Stress Caused by AI-Mediated Social Rejection"** clearly represents the core idea of the project. It focuses on understanding how users emotionally react when AI systems suddenly stop responding or become unavailable during interaction. The term **Digital Ghosting** refers to situations where artificial intelligence creates a feeling of silence or rejection similar to being ignored in human communication. The phrase **Stress Analyzer** highlights the practical implementation of the project, where user responses are examined to estimate emotional stress levels. This project combines research concepts with a working software model to study the psychological impact of AI communication failure in a measurable and meaningful way.

4. Objectives:

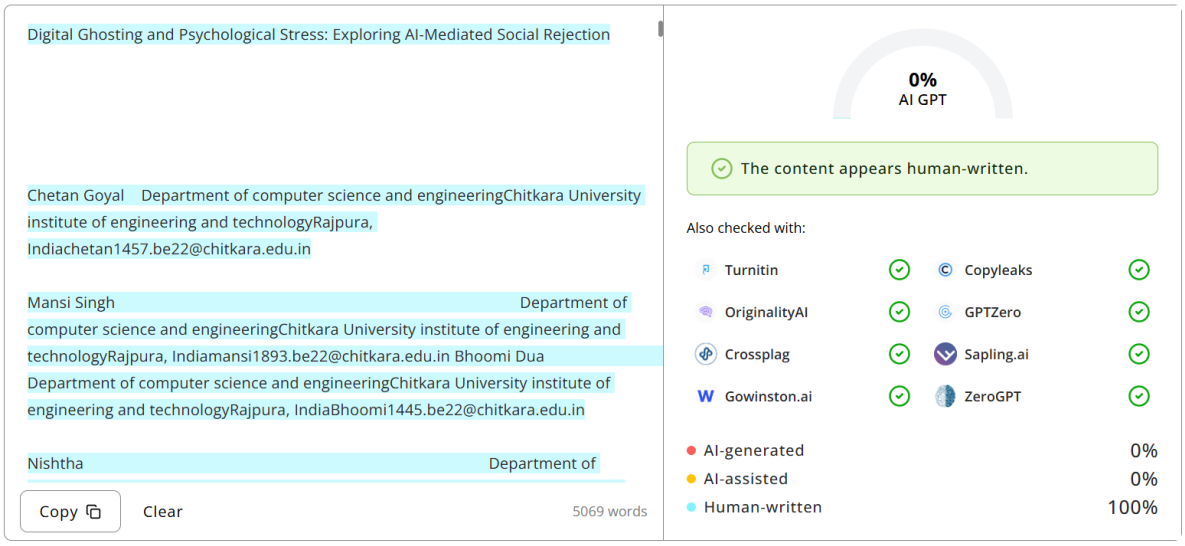
The main objective of this project is to analyze the psychological stress caused by sudden silence or non-responsiveness of AI systems using a practical and research-based approach. The specific objectives of the study are as follows:

1. To analyze the stress level of users when AI systems suddenly stop replying during communication or support-based interactions.
2. To classify user emotional responses into Low, Medium, and High stress categories based on text input.
3. To develop a working software prototype using Python and Streamlit for practical implementation of the research concept.
4. To present user stress data through charts and dashboard visualization for better understanding of behavioral patterns.
5. To study the growing dependence of users on AI communication tools and their emotional reactions during system failure.
6. To identify future improvements that can make AI systems more transparent, responsive, and user-friendly during downtime.

5. Options Available to Execute the Project:

- **Platform Used:** Visual Studio Code / PyCharm / Jupyter Notebook / Anaconda / Command Prompt / Terminal
- **Required Libraries:** streamlit, pandas, matplotlib
- **Language Used:** Python
- **Requirement:** Python 3.10 or above
- **Operating System:** Windows / macOS / Linux
- **Execution Command:** python -m streamlit run app.py

6. Similarity Score



7. Future Scope

The future scope of this project is wide, as the current version is a basic prototype designed to demonstrate the research concept in a practical manner. In upcoming versions, the system can be enhanced by integrating machine learning and natural language processing techniques for more accurate stress detection based on user responses. A secure login system and database connectivity can be added to store real-time user feedback and survey results. The dashboard can be improved with live analytics, downloadable reports, and comparative stress trends. The project may also be converted into a mobile application for easier accessibility. In addition, AI chatbots can be integrated with emotional support features and transparent error messages to reduce user frustration during service interruptions. This project can further contribute to research in human-computer interaction, digital psychology, and emotionally responsible AI systems.

8. Conclusion

The Digital Ghosting Stress Analyzer successfully transforms a research idea into a practical software application. It shows that emotional stress caused by AI silence can be studied through a simple and interactive system.

The project also highlights that technology should not only be efficient but emotionally responsible. As people increasingly depend on AI communication tools, understanding user well-being becomes important.

This prototype serves as a useful academic model that connects research findings with real-world implementation. It also creates scope for future intelligent mental wellness tools.

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